

Agnostic COT Prompt

Classify the business case given in library file according to the classification file and set in the format given by the brainstorm rules file. Set it up with all 9 categories as given by the brainstorm rules file. If there are any many to many classes relation, they form an association class between them, and they need any attributes necessary to make the association work. If any idea is not connected to any other idea by at least one connection, it is likely a false positive and should not be included. Only information that can stand as a quality of another or independently should be recorded at all.

Attributes can be scalar or strings, they don't need to be specifically numbers.

Any ideas that are appended to another are typically attributes of a that class.

Clarify that intermediate linking actions or transaction concepts should only be extracted if they explicitly hold unique properties or independent transaction records; otherwise, express them directly as binary association relations.

Instruct the model to strictly perform a secondary pass scanning for secondary structural nouns (e.g., sub-activities, structural facilities, or submitted materials) that host attributes or relationships, ensuring no functional entity mentioned in the text is omitted.

- **Precision: 92.59%**
- **Recall: 89.29%**
- **F1 Score: 90.91%**

Library Result

Library COT Result

Here is the step-by-step extraction and classification of the domain elements from the library business case (library - Tagged(1).pdf) according to the 9 categories specified in the brainstorm rules (Brainstorm Rules - Tagged.pdf) and classification guidelines (Classification Rules - Tagged-1.pdf).

Step 1: Brainstorming & 9-Category Classification

1. Nouns / Noun Phrases

- **Library** → (c) **Library** (Class)
- **Customer / Member** → (c) **Customer** (Class)
- **Membership Card** → (c) **Membership Card** (Class / Structural Facility holding attributes)
- **Member Number** → (a) **member number** (Attribute of Membership Card / Customer)
- **Name** → (a) **name** (Attribute of Customer)
- **Address** → (a) **address** (Attribute of Customer)
- **Date of Birth** → (a) **date of birth** (Attribute of Customer)
- **Subject Section** → (c) **Subject Section** (Class)
- **Classification Mark** → (a) **classification mark** (Attribute of Subject Section)
- **Loan Item / Item** → (c) **Loan Item** (Class)
- **Bar Code** → (a) **bar code** (Attribute of Loan Item)

- **Language Tape** → (c) **Language Tape** (Class)
- **Book** → (c) **Book** (Class)
- **Title** → (a) **title** (Attribute of Loan Item / Book / Language Tape)
- **Author(s)** → (a) **author(s)** (Attribute of Book)
- **Language** → (a) **language** (Attribute of Language Tape)
- **Level** → (a) **level** (Attribute of Language Tape)
- **Bar Code Reader** → (c) **Bar Code Reader** (Secondary Structural Facility hosting interaction)

2. "X of Y" Expressions

- **Name of customer** → `name` is an attribute of **Customer**
- **Address of customer** → `address` is an attribute of **Customer**
- **Date of birth of customer** → `date of birth` is an attribute of **Customer**
- **Classification mark of section** → `classification mark` is an attribute of **Subject Section**
- **Bar code of item / book** → `bar code` is an attribute of **Loan Item**
- **Level of tape** → `level` is an attribute of **Language Tape**
- **Title of tape / book** → `title` is an attribute of **Loan Item**
- **Author(s) of book** → `author(s)` is an attribute of **Book**
- **Number of items on loan** → `number of items on loan` is a dynamic derived attribute / constraint of **Customer**

3. Transitive Verbs (Association Relationships)

- **Borrow** (Customer borrows Loan Item) → **Association** between **Customer** and **Loan Item**. *(Note: Since a Customer can borrow multiple items and an Item can be borrowed by multiple customers over time, this forms a Many-to-Many relationship. Because intermediate transactional actions like borrowing/renewing do not explicitly hold unique independent transaction records specified in the text, it is modeled via an **Association Class: Loan / Borrow Record** to manage dates and status like borrowed, renewed, reserved).*
- **Issue** (Library issues Loan Item to Customer) → **Association**
- **Reserve** (Customer reserves Loan Item) → **Association** / Status state on **Loan / Borrow Record**
- **Renew** (Customer renews Loan Item) → **Association** / Action updating **Loan / Borrow Record**
- **Scan / Read** (Bar Code Reader scans Membership Card / Book Bar Code) → **Association** between **Bar Code Reader** and **Membership Card / Loan Item**
- **Search** (User searches Loan Item) → Action / Method on **Loan Item**

4. Adjectives / Enumerations

- **Valid / Invalid** (Membership validity) → Enumerated attribute value (`validityStatus`) of **Membership Card / Customer**
- **Beginner / Intermediate / Advanced** (Tape level) → Attribute value for `level` in **Language Tape**
- **Borrowed / Reserved / Renewed** → Enumerated attribute value (`loanStatus`) of **Loan / Borrow Record**

5. Numeric / Quantity

- **8** (Maximum items borrowed) → Multiplicity constraint on **Customer borrowing Loan Item** (0..8)
- **1** (Unique member number, unique barcode) → Multiplicity / Key attribute constraint

6. Possession Expressions ("has/have, possess")

- **Customer has Membership Card** → Association / Aggregation between **Customer** and **Membership Card**
- **Language tape has title, language, level** → Attributes of **Language Tape**
- **Book has title, author(s)** → Attributes of **Book**

7. "Consist of / Part of" Expressions (Aggregation / Composition)

- **Library is made up of Subject Sections** → Aggregation / Composition (**Library** ◇--- **Subject Section**)
- **Subject Section contains Loan Items** → Aggregation (**Subject Section** ◇--- **Loan Item**)

8. Containment / Containing Expressions

- **Library contains Subject Sections** → Aggregation
- **Subject Section contains Loan Items** → Aggregation

9. "X is a Y" Expressions (Generalization / Inheritance)

- **Language Tape is a Loan Item** → Inheritance (**Language Tape** → **Loan Item**)
- **Book is a Loan Item** → Inheritance (**Book** → **Loan Item**)
- **Member is a Customer** → Synonymous / Role refinement (**Customer**)

Step 2: Secondary Pass Structural Entity Verification

A secondary pass ensures no structural facilities or secondary sub-activities holding attributes or relationships are omitted:

1. **Membership Card**: Structural facility issued to a customer that holds `memberNumber` and `validityStatus`.
2. **Bar Code Reader**: Structural physical facility used to scan `Membership Card` and `Loan Item`.
3. **Loan / Borrow Record (Association Class)**: Created to resolve the Many-to-Many relation between **Customer** and **Loan Item**. It carries the minimal necessary properties to make the association work (`issueDate`, `dueDate`, `status`: borrowed/reserved/renewed).

Summary of Extracted Structural Model

Classes & Attributes

- **Customer / Member**
 - `name` : String
 - `address` : String
 - `dateOfBirth` : String / Date
 - `currentLoansCount` : Integer (Max 8)
- **Membership Card**
 - `memberNumber` : String / Identifier

- `isValid` : Boolean
- **Subject Section**
- `classificationMark` : String
- **Loan Item** (Abstract Superclass)
- `barCode` : String / Identifier
- `title` : String
- `isReserved` : Boolean
- **Book** (Subclass of Loan Item)
- `author` : String
- **Language Tape** (Subclass of Loan Item)
- `language` : String
- `level` : String
- **Bar Code Reader** (Hardware / Structural Facility)
- `readerID` : String

Association Class

- **LoanRecord** (Association Class between **Customer** and **Loan Item**)
- `issueDate` : Date
- `dueDate` : Date
- `status` : Enum {Borrowed, Reserved, Renewed}

Relationships

1. **Inheritance (IS-A)**: Book IS-A Loan Item, Language Tape IS-A Loan Item.
2. **Aggregation (PART-OF)**: Library consists of Subject Section; Subject Section contains Loan Item.
3. **Association / Association Class**: Customer borrows Loan Item (0..8 items per customer) linked via LoanRecord.
4. **Binary Association**: Customer holds Membership Card; Bar Code Reader reads Membership Card and Loan Item.

Car rental Result

Car rental Result of COT

Here is the classification and analysis of the business case from the **Car Rental Application**, structured according to the **Brainstorming Rules** and **Classification Rules**.

1. Nouns / Noun Phrases (Classes & Attributes)

- **Rental Location** (Class)
- **Vehicle / Car** (Class, generalized from passenger cars)
- **Manufacturer** (Class)
- **Make** (Class / Attribute)
- **Model** (Class / Attribute)
- **Price Class** (Class)

- **Customer** (Class)
- **Rental Plan** (Class)
- **Salesperson** (Class)
- **Reservation Form / Reservation** (Class / Transaction Concept with independent records)
- **Invoice** (Class / Transaction Concept with independent records)
- **Company / Employer** (Class)
- **Credit Card Processing Company** (Class)
- **Maintenance / Repair Record** (Association Class / Transaction Concept)

2. "X of Y" Expressions

- **Costs of running its business** → Attribute of Business Operation / Rental Company.
- **Labor cost** → Attribute of Business Operation.
- **Make of car** → Association/Attribute linking Car to Make.
- **Model of car** → Association/Attribute linking Car to Model.
- **Rental price** → Attribute of Rental Plan / Option.
- **Employer of customer** → Association relationship between Customer and Company.

3. Transitive Verbs

- **Operate** (rental location operates throughout the metropolitan area)
- **Rent** (customer rents a car)
- **Select** (customer selects make and model)
- **Display** (system displays a message)
- **Suggest** (system suggests similar models)
- **Process** (salesperson processes reservations manually)
- **Archive** (salesperson archives reservations in the file cabinet)
- **Honor** (reservation is honored if cars are available)
- **Check out** (car is checked out to a customer, opening an invoice)
- **Settle** (customer settles the invoice)
- **Repair / Maintain** (cars need preventive maintenance and repair)

4. Adjectives, Enumeration (Options & States)

- **Passenger cars** (Enumeration/Specialization of vehicles)
- **Automatic or manual** (gear change options)
- **Two or four** (door options)
- **Sedan or hatchback** (body style options)
- **Available / Rented out** (Car availability state)
- **Daily unlimited miles plan / Weekend 10% discount plan** (Rental plan enumerations)

5. Numeric, Quantity

- **Several years** (Time duration for business growth/cost increase)
- **One or more rentals** (Multiplicity covered by a single invoice)
- **Given period of time** (Reservation validity threshold window)
- **Depreciation** (Numeric/financial valuation attribute for tax purposes)

6. Possession Expressions (Has / Have / Possess)

- **Company has a rental fleet / rental locations** → Aggregation / Association.
- **Make has several models** → Association / Aggregation relationship.
- **Car has options** (gear change, doors, body style) → Attributes of Car.
- **Invoice has rentals** → Association relationship.

7. "Consist of / Part of" Expressions

- **Models are grouped into price classes** → Classification / Aggregation hierarchy.
- **Rental fleet consists of different makes and models of cars** → Aggregation relationship.

8. Containment / Containing Expressions

- **File cabinet containing archived reservation forms** → Physical containment of transaction records.
- **Rental location containing rental cars throughout the metropolitan area** → Spatial containment / aggregation.

9. "X is a Y" Expressions (Generalization / Specialization)

- **Passenger car is a vehicle** → Inheritance / Specialization relationship.
- **Company (Employer) is a company** → Generalization.

Association Classes & Secondary Structural Analysis

- **Block Reservation / Multi-Car Rental Association:** Because a single invoice can cover one or more rentals and block reservations handle multiple cars together, an association class (`Rental Transaction`) links **Customer**, **Car**, and **Invoice** to capture independent transaction records, dates, and charges.
- **Maintenance & Repair Association:** To track purchase, repair, maintenance, and disposal information for depreciation and tax purposes, a maintenance log association class links **Car** with service attributes (repair cost, maintenance type, date).

Note on False Positives: Intermediate actions or general descriptors without independent existence or unique properties (such as general conversational phrasing like "business has boomed") have been excluded in accordance with the filtering rules.

NTSS Result

NTSS COT Result

Based on the domain details provided in **ntss-new-1.pdf** and structured according to the classification rules in **Brainstorm Rules - Tagged.pdf** and **Classification Rules - Tagged-1.pdf**, here is the 9-category domain analysis mapping.

1. Noun / Noun Phrase (Class or Attribute)

- **Business Customer / Customer:** Class (independent entity ordering NTSS services)
- **NTSS (National Trade Show Services):** Class (service provider entity)
- **Trade Show / Event:** Class (core domain concept holding schedule, venue, etc.)
- **Account:** Class (tracks financial charges and balances for each customer)

- **Organizer:** Class (person or organization organizing the event)
- **Participant:** Class (base entity for people attending the event)
- **NTSS Staff:** Class (sub-class of participant/staff that manages events)
- **Invited Speaker:** Class (speaker invited for keynote address)
- **Selected Speaker:** Class (speaker selected via proposal process)
- **Reviewer / Reviewer Committee:** Class (domain experts who evaluate proposals)
- **Exhibitor:** Class (entity renting booths to show products/services)
- **Observer:** Class (attendee visiting trade show)
- **Proposal:** Class / Entity (submitted material with attributes like status and content)
- **Booth:** Class / Facility (structural facility with size attributes and rental contracts)
- **Seminar Room / Conference Room:** Class / Facility (structural facility host for seminars)
- **Predefined Domain:** Class (classification category for trade shows)
- **Service:** Class (professional service offered by NTSS)

2. "X of Y" Expression (X is an Attribute of Y)

- **Design of Trade Show:** Theme, Slogan, Location, Duration → attributes of Trade Show
- **Location of Event:** Address / Center (e.g., Las Vegas Convention Center, Moscone Center) → attribute of Trade Show
- **Duration of Event:** Start date, End date (e.g., January 7-10) → attribute of Trade Show
- **Contact Information of Event:** Phone/Email → attribute of Trade Show / Organizer
- **Website of Event:** URL string → attribute of Trade Show
- **Status of Proposal:** Pending review, Accepted, Rejected → attribute of Proposal
- **Size of Booth:** Large, Medium, Small → attribute of Booth
- **Service Charges / Payments Received / Account Balance of Account:** Numerical/currency values → attributes of Account

3. Transitive Verbs (Association Relationships / Association Classes)

- **Customer contacts NTSS / orders Service:** Direct association between Customer and Service / NTSS
- **NTSS creates Account for Customer:** Association between NTSS, Customer, and Account
- **Organizer organizes Event:** Association between Organizer and Trade Show
- **Reviewer reviews Proposal:** Many-to-Many association forming ****Proposal Review**** (Association Class) with evaluation score/comments.
- **Participant registers for Event:** Many-to-Many association forming ****Event Registration**** (Association Class) storing unique properties (registrationFee, registrationDate, badgeNumber).
- **Exhibitor rents Booth:** Many-to-Many association forming ****Booth Rental**** (Association Class) storing rental specific details (rentalFee, duration, boothNumber).

4. Adjectives / Enumerations (Attribute Values)

- **Proposal Status:** PENDING_REVIEW, ACCEPTED, REJECTED
- **Booth Size:** LARGE, MEDIUM, SMALL
- **Organizer Type:** PERSON, ORGANIZATION
- **Trade Show Sub-Activities:** CREATION, PROMOTION, ORGANIZATION, RUNNING

5. Numeric / Quantity (Attribute Values or Multiplicity Values)

- **Duration:** Number of days
- **Registration Fee:** Currency scalar (\$)
- **Booth Rental Fee:** Currency scalar (\$)
- **Account Balance / Charges:** Currency scalar (\$)
- **Multiplicities:** 1 Customer to 1 Account; 1 Event to 1.. Predefined Domains; 1 Exhibitor to 1.. Booths.

6. Possession Expressions (Aggregation, Association, or Attribute)

- **Customer has Account:** Binary association
- **Event has Contact Info & Website:** Direct scalar attributes of Trade Show / Event
- **Booth has Size:** Scalar attribute of Booth
- **Proposal has Status:** Enumerated attribute of Proposal

7. "Consist of / Part of" Expressions (Aggregation Relationships)

- **Trade Show consists of Sub-Activities:** Design/Creation, Promotion, Organization, Running
- **Trade Show consists of Structural Facilities:** Booths, Conference/Seminar Rooms
- **Trade Show consists of Trade Show Materials:** Physical distributed items

8. Containment / Containing Expressions (Association or Aggregation)

- **Event belongs to / contained in Predefined Domain:** Aggregation/Association between Trade Show and Predefined Domain
- **Conference Room contains Seminars:** Structural containment relation

9. "X is a Y" Expressions (Inheritance Relationships)

- **Trade Show is an Event** (Trade Show $\rightarrow$ Event)
- **NTSS Staff is a Participant** (NTSS Staff $\rightarrow$ Participant)
- **Speaker is a Participant** (Speaker $\rightarrow$ Participant)
- **Invited Speaker is a Speaker** (Invited Speaker $\rightarrow$ Speaker)
- **Selected Speaker is a Speaker** (Selected Speaker $\rightarrow$ Speaker)
- **Exhibitor is a Participant** (Exhibitor $\rightarrow$ Participant)
- **Observer is a Participant** (Observer $\rightarrow$ Participant)